PROBLEM 26

Heavy Particle on a Rotating Half-Circumference (Moving-Frame Method)

Original statement · Problems.pdf, p. 14

Rotating Wire + Inertia Forces

A heavy point particle P of mass m is constrained to move without friction along the half-circumference wire of radius a shown in the figure. The wire containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

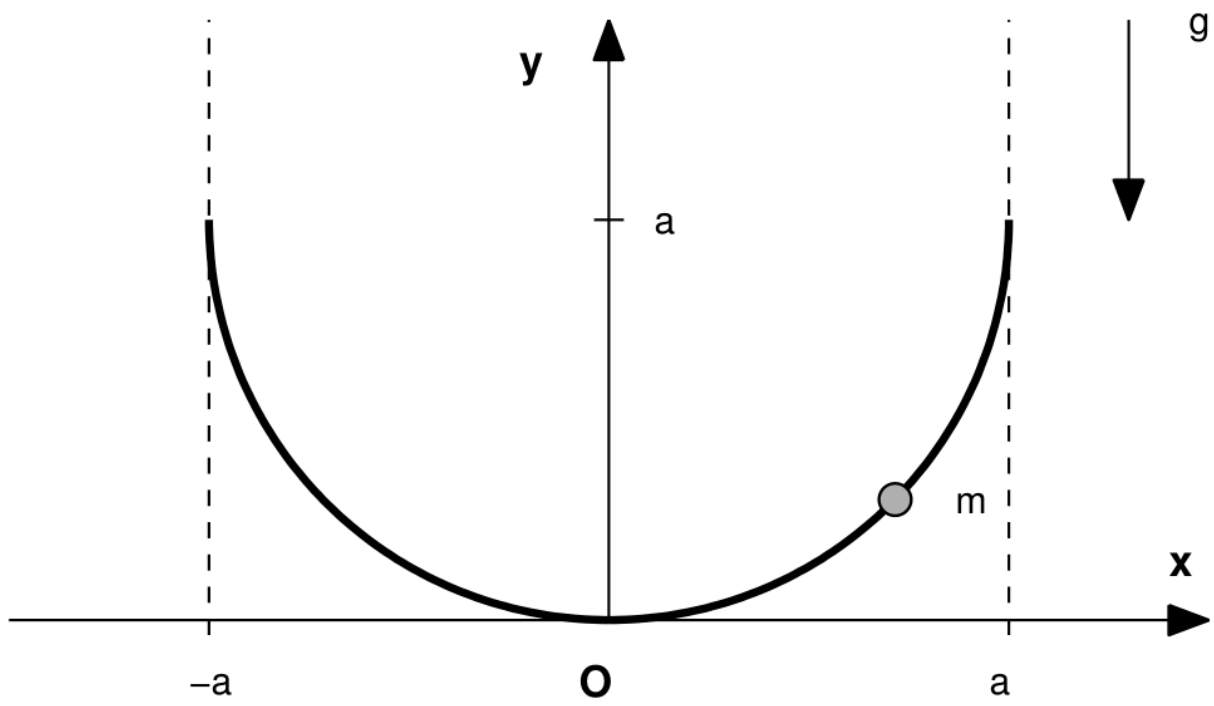


Figure: Half-circumference wire of radius a with lowest point O, ends at x = ±a, rotating about the vertical axis OY; particle of mass m on the wire (Problems.pdf, p. 14).

(a) Determine the number of degrees of freedom of the problem.

(b) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Express the inertia force that acts on the particle in this moving reference frame.

(c) Express also the components of the weight of the particle with respect to $OX_1Y_1Z_1$.

(d) Determine if there are any equilibrium points.

(e) Reduce the problem to integrals (i.e. quadratures) and determine the value of the reaction force.

(f) Is it possible to use the energy equation? If Ω were a function of time, would it be possible to use the energy equation? Justify your answer.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a heavy particle P (mass m) slides without friction on a half-circumference wire of radius a. In the plane of the wire the lowest point is O, the centre is $C = (0, a)$ and the ends are $(\pm a, a)$ (figure of the statement). The wire turns about the vertical axis OY with constant angular velocity $\boldsymbol{\omega} = \Omega\,\mathbf{j}$.
- **Hypotheses:** the frame $OXYZ$ with OY vertical upwards is inertial; gravity is $-g\,\mathbf{j}$; the wire is smooth, so its reaction has no component along the wire; the rotation is imposed (whatever torque is needed is supplied). It is solved with the relative-motion method of Topic 5 (moving frame, Notes Eq. 3.27 and inertia forces, Notes Eqs. 4.62–4.63).
- **Why a moving frame:** in a frame that turns with the wire the constraint is a fixed circle, so the motion of P is a one-dimensional motion on a fixed curve. The cost is to add the centrifugal and Coriolis forces; there is no Euler force and no force from the acceleration of the origin because Ω is constant and O is fixed.
- **Initial conditions ($t = 0$):** P at O ($x = 0$) with velocity v_0 along $+x$.
- **Fixed (inertial) frame:** $S_0 = OXYZ = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$, with $\mathbf{j}$ vertical upwards.
- **Moving frame** (requested in (b)): $S_1 = OX_1Y_1Z_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the wire: OX_1 in the plane of the wire (the x axis of the figure), $OY_1 \equiv OY$ (the axis of rotation) and OZ_1 normal to the plane of the wire. S_1 coincides with S_0 at $t = 0$.

- **Intrinsic basis of the wire:** $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ at P: unit tangent in the sense of increasing θ, principal normal and binormal. They are deduced in the next section.
- **Coordinate:** θ, the angle of CP measured at the centre C from the downward vertical CO, positive towards $+X_1$, with $\theta \in [-\pi/2, \pi/2]$ on the wire. Ωt is prescribed, so it is a known function of time, not a coordinate.

Symbol	Meaning	SI unit
m	mass of P	kg
a	radius of the wire	m
Ω	angular velocity of the wire (constant)	rad/s
g	gravitational acceleration	m/s^2
v_0	initial speed at O	m/s
θ	angle of CP from the downward vertical	rad

(a) Degrees of freedom

> **Pedagogical Justification:** CDOF = (coordinates of the free particle) − (independent holonomic constraints).

P has 3 coordinates. In S_1 the wire imposes two fixed equations, $z_1 = 0$ (plane of the wire) and $x_1^2 + (y_1 - a)^2 = a^2$ (circle of centre C). In S_0 the same constraints are rheonomic (they turn with Ωt), but the count is the same: CDOF $= 3 - 2 = 1$, with generalized coordinate θ.

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

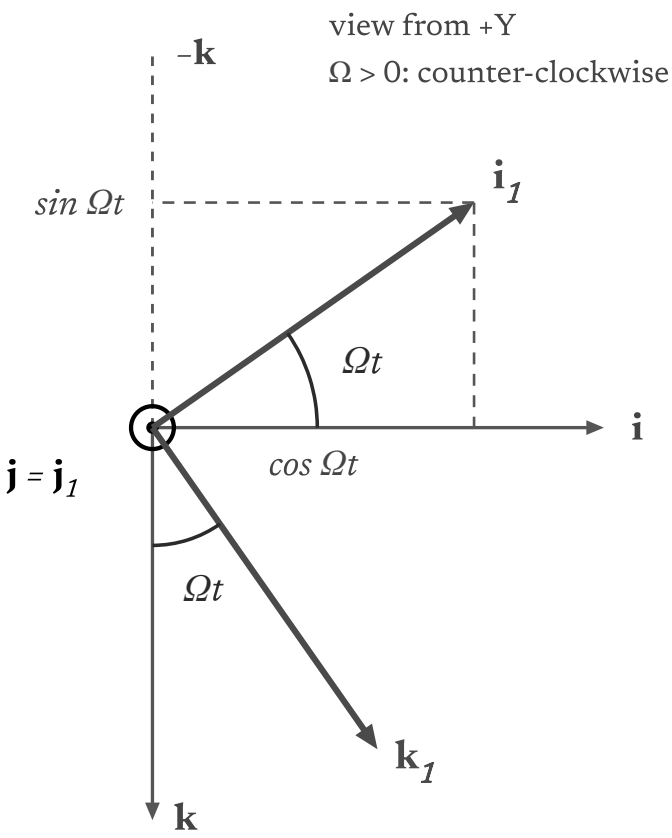


Figure 1. View from $+Y$, with $\mathbf{j} = \mathbf{j}_1$ out of the paper ($\odot$): then $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ points down the page. $\mathbf{i}_1, \mathbf{k}_1$ are $\mathbf{i}, \mathbf{k}$ turned counter-clockwise by Ωt; the components of $\mathbf{i}_1$ are $\cos \Omega t$ along $\mathbf{i}$ and $\sin \Omega t$ along $-\mathbf{k}$ (up the page), hence the minus sign.

The rotation is about $\mathbf{j}$, so it is drawn looking down the axis, from $+Y$. In that view $\mathbf{j}$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}$ to the right. The direction of $\mathbf{k}$ then follows from the right-hand rule, $\mathbf{k} = \mathbf{i} \times \mathbf{j}$: the fingers go from $\mathbf{i}$ (right) to $\mathbf{j}$ (towards the reader) and the thumb points down the page. So in this view $\mathbf{k}$ points down and $-\mathbf{k}$ points up.

With $\boldsymbol{\omega} = \Omega\,\mathbf{j}$ and $\Omega > 0$, the wire turns counter-clockwise in this view (right-hand rule again: thumb along $\mathbf{j}$, towards the reader). After a time t the axis $\mathbf{i}_1$ has turned the angle Ωt from $\mathbf{i}$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the right triangle drawn with dashed lines. The leg adjacent to Ωt lies along $\mathbf{i}$ and measures $\cos \Omega t$. The leg opposite to Ωt points up the page, which is the direction $-\mathbf{k}$, and measures $\sin \Omega t$. Hence $\mathbf{i}_1 = \cos \Omega t\, \mathbf{i} + \sin \Omega t\, (-\mathbf{k})$. **The**

minus sign comes from k pointing down the page in this view.

- $\mathbf{k}_1$ starts on $\mathbf{k}$ (down the page) and turns the same angle Ωt counter-clockwise, which moves its tip towards $+\mathbf{i}$. Its leg along $\mathbf{k}$ (adjacent) is $\cos\Omega t$ and its leg along $\mathbf{i}$ (opposite) is $+\sin\Omega t$.
- $\mathbf{j}_1 = \mathbf{j}$, because it is the axis of the rotation.

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}, \qquad \mathbf{j}_1 = \mathbf{j}, \qquad \mathbf{k}_1 = \sin\Omega t\,\mathbf{i} + \cos\Omega t\,\mathbf{k}$$

The columns of the change-of-basis matrix are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$ (vault note *Topic 5 - Relative Motion*, Sec. 4.1):

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\Omega t\cos\Omega t - \sin\Omega t\,(-\sin\Omega t) = \cos^2\Omega t + \sin^2\Omega t = 1$$

Each column has unit length ($\cos^2\Omega t + \sin^2\Omega t = 1$) and the first and third are orthogonal ($\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$; also $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1$, so $\mathcal{B}_1$ is right-handed.

Motion of S_1. $\mathbf{r}_0^O = \mathbf{0}$, $\mathbf{v}_0^O = \mathbf{0}$, $\mathbf{a}_0^O = \mathbf{0}$ (fixed origin); $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_1$ and $\boldsymbol{\alpha}_{10} = \dot{\Omega}\,\mathbf{j}_1 = \mathbf{0}$. Check with Poisson's formula: by the chain rule, $\frac{d\mathbf{i}_1}{dt}\big|_0 = \frac{d(\cos\Omega t)}{dt}\,\mathbf{i} - \frac{d(\sin\Omega t)}{dt}\,\mathbf{k} = -\Omega\sin\Omega t\,\mathbf{i} - \Omega\cos\Omega t\,\mathbf{k} = -\Omega\,\mathbf{k}_1$, and indeed $\boldsymbol{\omega}_{10} \times \mathbf{i}_1 = \Omega\,\mathbf{j}_1 \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.

The position vector in the plane of the wire

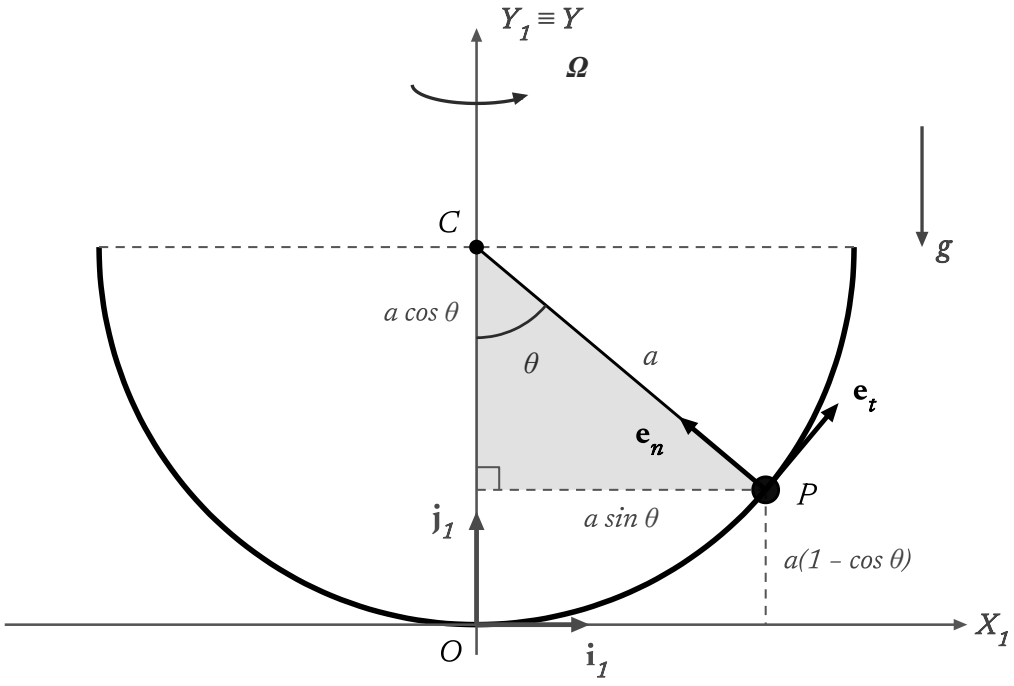


Figure 2. Plane of the wire, fixed in S_1: θ is measured at C from the downward vertical CO. In the shaded right triangle the hypotenuse is $|CP| = a$, the leg opposite θ is $a\sin\theta$ and the adjacent leg is $a\cos\theta$; the dashed lines are the components of $\mathbf{r}_1^P$. $\mathbf{i}_1, \mathbf{j}_1$ in blue, $\mathbf{e}_t, \mathbf{e}_n$ in orange.

The plane of the wire is fixed in S_1, so it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up. To go from O to P we pass through the centre, because the distance from C is the one that stays constant:

$$\mathbf{r}_1^P = \overrightarrow{OC} + \overrightarrow{CP}$$

- $\overrightarrow{OC} = a\,\mathbf{j}_1$: C is on the axis, at height a above O.
- $\overrightarrow{CP}$ has length a and makes the angle θ with the downward vertical $-\mathbf{j}_1$. Let F be the point of the axis at the height of P. In the shaded right triangle CFP (right angle at F) the hypotenuse is $|CP| = a$; the leg opposite to θ is $FP = a\sin\theta$, horizontal and towards

$+\mathbf{i}_1$ for $\theta > 0$; the leg adjacent to θ is $CF = a\cos\theta$, vertical and pointing down, along $-\mathbf{j}_1$. So $\overrightarrow{CP} = a\sin\theta\,\mathbf{i}_1 - a\cos\theta\,\mathbf{j}_1$.

Adding the two vectors:

$$\mathbf{r}_1^P = a\sin\theta\,\mathbf{i}_1 + a\,(1-\cos\theta)\,\mathbf{j}_1$$

These are the two dashed components of the figure: the horizontal one $a\sin\theta$ and the height of P above O, $a - a\cos\theta$. Checks: $\theta = 0$ gives $\mathbf{r}_1^P = \mathbf{0}$ (the particle at O, the initial position); $\theta = \pi/2$ gives $a\,\mathbf{i}_1 + a\,\mathbf{j}_1$, the end (a, a) of the wire.

Relation with the standard polar angle. The usual polar angle φ about C is measured from $\mathbf{i}_1$, counter-clockwise, with $\overrightarrow{CP} = a\,(\cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1)$. Comparing with $\overrightarrow{CP} = a\sin\theta\,\mathbf{i}_1 - a\cos\theta\,\mathbf{j}_1$ requires $\cos\varphi = \sin\theta$ and $\sin\varphi = -\cos\theta$, which holds for

$$\varphi = \theta - \frac{\pi}{2}: \qquad \cos\left(\theta - \tfrac{\pi}{2}\right) = \cos\theta\cos\tfrac{\pi}{2} + \sin\theta\sin\tfrac{\pi}{2} = \sin\theta, \qquad \sin\left(\theta - \tfrac{\pi}{2}\right) = \sin\theta\cos\tfrac{\pi}{2} - \cos\theta\sin\tfrac{\pi}{2} = -\cos\theta$$

(with $\cos\frac{\pi}{2} = 0$ and $\sin\frac{\pi}{2} = 1$). So θ is the polar angle measured from $-\mathbf{j}_1$ instead of $\mathbf{i}_1$. It is chosen so that $\theta = 0$ is the lowest point O, where the particle starts and where the trivial equilibrium is.

Intrinsic basis of the wire. The tangent is the derivative of the position with respect to the coordinate. With $\frac{d}{d\theta}\sin\theta = \cos\theta$ and $\frac{d}{d\theta}(1-\cos\theta) = \sin\theta$:

$$\frac{d\mathbf{r}_1^P}{d\theta} = a\cos\theta\,\mathbf{i}_1 + a\sin\theta\,\mathbf{j}_1, \qquad \left|\frac{d\mathbf{r}_1^P}{d\theta}\right| = a\sqrt{\cos^2\theta + \sin^2\theta} = a \;\Rightarrow\; \mathbf{e}_t = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1$$

Differentiating $\mathbf{e}_t$ with respect to θ ($\frac{d}{d\theta}\cos\theta = -\sin\theta$, $\frac{d}{d\theta}\sin\theta = \cos\theta$) gives a vector of length $\sqrt{\sin^2\theta + \cos^2\theta} = 1$, the principal normal:

$$\mathbf{e}_n = \frac{d\mathbf{e}_t}{d\theta} = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1 = -\frac{\overrightarrow{CP}}{a}$$

so $\mathbf{e}_n$ points from P towards the centre C, as drawn. The binormal follows from $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1$ and $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1$:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \cos^2\theta\,(\mathbf{i}_1 \times \mathbf{j}_1) - \sin^2\theta\,(\mathbf{j}_1 \times \mathbf{i}_1) = \left(\cos^2\theta + \sin^2\theta\right)\mathbf{k}_1 = \mathbf{k}_1$$

The matrix with columns $\mathbf{e}_t, \mathbf{e}_n, \mathbf{k}_1$ in $\mathcal{B}_1$ is a rotation by θ about $\mathbf{k}_1$, with determinant $\cos^2\theta + \sin^2\theta = 1$. In the polar language, $\mathbf{u}_\rho = -\mathbf{e}_n$ (radial, from C outwards) and $\mathbf{u}_\varphi = \mathbf{e}_t$.

Kinematics

(b) Relative velocity in $OX_1Y_1Z_1$

In S_1 the vectors $\mathbf{i}_1, \mathbf{j}_1$ are constant, so only the components of $\mathbf{r}_1^P$ change. By the chain rule, $\frac{d}{dt}\sin\theta = \cos\theta\,\dot\theta$ and $\frac{d}{dt}(1-\cos\theta) = \sin\theta\,\dot\theta$, that is $\mathbf{v}_1^P = \frac{d\mathbf{r}_1^P}{d\theta}\,\dot\theta$:

$$\mathbf{v}_1^P = a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1 = a\dot\theta\,\mathbf{e}_t$$

Relative acceleration. Product rule on $a\dot\theta\,\mathbf{e}_t$ in S_1: $\frac{d}{dt}(a\dot\theta) = a\ddot\theta$ and, by the chain rule, $\frac{d\mathbf{e}_t}{dt}\big|_1 = \frac{d\mathbf{e}_t}{d\theta}\,\dot\theta = \dot\theta\,\mathbf{e}_n$:

$$\mathbf{a}_1^P = a\ddot\theta\,\mathbf{e}_t + a\dot\theta\,\dot\theta\,\mathbf{e}_n = a\ddot\theta\,\mathbf{e}_t + a\dot\theta^2\,\mathbf{e}_n$$

It is the circular motion of radius a: a tangential part and a centripetal part towards C.

Forces and Equations of Motion

(b, continued) Inertia forces

> **Pedagogical Justification:** Notes Eq. 4.62 (Slide 28) gives $\mathbf{F}_i = -m\mathbf{a}_0^O - m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P - m\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) - 2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$. The first two terms vanish ($\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\alpha}_{10} = \mathbf{0}$).

Centrifugal force. With $\mathbf{j}_1\times\mathbf{i}_1 = -\mathbf{k}_1$ and $\mathbf{j}_1\times\mathbf{j}_1 = \mathbf{0}$, $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \Omega\,\mathbf{j}_1\times a\sin\theta\,\mathbf{i}_1 = -\Omega a\sin\theta\,\mathbf{k}_1$; then, with $\mathbf{j}_1\times\mathbf{k}_1 = \mathbf{i}_1$, $\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) = \Omega\,\mathbf{j}_1\times(-\Omega a\sin\theta\,\mathbf{k}_1) = -\Omega^2 a\sin\theta\,\mathbf{i}_1$:

$$\mathbf{F}_C = m\Omega^2 a \sin\theta\, \mathbf{i}_1$$

It points horizontally away from the axis and equals $m\Omega^2 \mathbf{r}_\perp$ (Slide 30), since $\mathbf{r}_\perp = a\sin\theta\,\mathbf{i}_1$ is the part of $\mathbf{r}_1^P$ perpendicular to $\mathbf{j}_1$ (the horizontal leg of the triangle).

Coriolis force. $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \Omega\,\mathbf{j}_1 \times \left(a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1\right) = a\Omega\dot\theta\cos\theta\,(\mathbf{j}_1\times\mathbf{i}_1) = -\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = 2m\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$$

Total inertia force:

$$\mathbf{F}_i = m\Omega^2 a\sin\theta\,\mathbf{i}_1 + 2m\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$$

Its projections on the wire basis use $\mathbf{i}_1\cdot\mathbf{e}_t = \cos\theta$, $\mathbf{i}_1\cdot\mathbf{e}_n = -\sin\theta$ and $\mathbf{k}_1\cdot\mathbf{e}_t = \mathbf{k}_1\cdot\mathbf{e}_n = 0$: $\mathbf{F}_C\cdot\mathbf{e}_t = m\Omega^2 a\sin\theta\cos\theta$ and $\mathbf{F}_C\cdot\mathbf{e}_n = -m\Omega^2 a\sin^2\theta$.

(c) Weight in $OX_1Y_1Z_1$

The axis of rotation is the vertical, so $\mathbf{j}_1 = \mathbf{j}$ at all times and

$$\mathbf{W} = -mg\,\mathbf{j}_1 \qquad (W_{X_1}=0,\quad W_{Y_1}=-mg,\quad W_{Z_1}=0)$$

In the wire basis, with $\mathbf{j}_1\cdot\mathbf{e}_t = \sin\theta$ and $\mathbf{j}_1\cdot\mathbf{e}_n = \cos\theta$: $\mathbf{W}\cdot\mathbf{e}_t = -mg\sin\theta$, $\mathbf{W}\cdot\mathbf{e}_n = -mg\cos\theta$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$.

Reaction of the wire. The wire is smooth, so the reaction has no tangential component: $\mathbf{N} = N_n\,\mathbf{e}_n + N_z\,\mathbf{k}_1$.

Equations of motion.

> **Pedagogical Justification:** Newton's law in S_1, $m\mathbf{a}_1^P = \mathbf{W}+\mathbf{N}+\mathbf{F}_i$ (Notes Eq. 4.63), projected on the intrinsic basis. The tangent $\mathbf{e}_t$ eliminates the smooth reaction ($\mathbf{N}\cdot\mathbf{e}_t = 0$) and the Coriolis force ($\mathbf{k}_1\cdot\mathbf{e}_t = 0$); the other two projections give the reaction.

Projection on $\mathbf{e}_t$ (the equation of motion):

$$ma\ddot{\theta} = -mg\sin\theta + m\Omega^2 a\sin\theta\cos\theta \;\Rightarrow\; \ddot{\theta} = \sin\theta\left(\Omega^2\cos\theta - \frac{g}{a}\right)$$

Projection on $\mathbf{e}_n$:

$$ma\dot{\theta}^2 = N_n - mg\cos\theta - m\Omega^2 a\sin^2\theta$$

Projection on $\mathbf{k}_1$ ($\mathbf{a}_1^P\cdot\mathbf{k}_1 = 0$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$):

$$0 = N_z + 2m\Omega a\dot{\theta}\cos\theta$$

Answers and Checks

(a) Degrees of freedom

CDOF $= 1$, with generalized coordinate θ; the rotation Ωt is prescribed.

(b) Velocity and inertia forces in $OX_1Y_1Z_1$

$\mathbf{v}_1^P = a\dot{\theta}\,\mathbf{e}_t = a\dot{\theta}\cos\theta\,\mathbf{i}_1 + a\dot{\theta}\sin\theta\,\mathbf{j}_1$ and $\mathbf{F}_i = m\Omega^2 a\sin\theta\,\mathbf{i}_1 + 2m\Omega a\dot{\theta}\cos\theta\,\mathbf{k}_1$.

(c) Weight

$\mathbf{W} = -mg\,\mathbf{j}_1$.

(d) Equilibrium points

Equilibria (relative rest, $\ddot{\theta} = \dot{\theta} = 0$, in the tangential equation):

1. $\sin\theta = 0$: $\theta = 0$, the lowest point O. It exists for every Ω.
2. $\cos\theta_* = \dfrac{g}{a\Omega^2}$: two symmetric points $\theta = \pm\theta_*$, which exist only if $\Omega^2 > g/a$ (then $0 < \theta_* < \pi/2$, inside the wire).

Stability with the potential of (f), $V_1(\theta) = mga(1-\cos\theta) - \frac{1}{2}m\Omega^2 a^2\sin^2\theta$ (the tangential equation is $ma^2\ddot{\theta} = -dV_1/d\theta$). Its derivatives are $V_1' = ma\sin\theta\left(g - \Omega^2 a\cos\theta\right)$ and $V_1'' = mga\cos\theta - m\Omega^2 a^2\cos 2\theta$:

- $V_1''(0) = ma\left(g - \Omega^2 a\right)$: $\theta = 0$ is stable if $\Omega^2 < g/a$ and unstable if $\Omega^2 > g/a$.

- $V_1''(\pm\theta_*) = mga\cos\theta_* - m\Omega^2a^2\left(2\cos^2\theta_* - 1\right) = m\Omega^2a^2\left(1-\cos^2\theta_*\right) = m\Omega^2a^2\sin^2\theta_* > 0$ (using $ga = \Omega^2a^2\cos\theta_*$): the off-axis equilibria are stable whenever they exist.

(e) Quadratures and reaction force

> **Pedagogical Justification:** multiplying the tangential equation by $\dot\theta$ makes both sides exact derivatives; the result is the energy integral of (f). The projections on $\mathbf{e}_n$ and $\mathbf{k}_1$ then give the reaction.

First integral. Multiply $ma^2\ddot\theta = -mga\sin\theta + m\Omega^2a^2\sin\theta\cos\theta$ by $\dot\theta$. By the chain rule, $\frac{d}{dt}\left(\frac12\dot\theta^2\right) = \dot\theta\ddot\theta$, $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot\theta$ and $\frac{d}{dt}\left(\frac12\sin^2\theta\right) = \sin\theta\cos\theta\,\dot\theta$, so

$$\frac{d}{dt}\left[\tfrac12 ma^2\dot\theta^2 - mga\cos\theta - \tfrac12 m\Omega^2a^2\sin^2\theta\right] = 0$$

Integrate from 0 to t and apply Barrow's rule, with $\theta(0) = 0$ and $a\dot\theta(0) = v_0$ (at O, $\mathbf{r}_1^P = \mathbf{0}$, so $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \mathbf{0}$ and the given absolute velocity equals the relative one, $v_0\,\mathbf{i}_1 = a\dot\theta(0)\,\mathbf{e}_t(0)$):

$$\left[\tfrac12 ma^2\dot\theta^2 - mga\cos\theta - \tfrac12 m\Omega^2a^2\sin^2\theta\right]_0^t = 0 \;\Rightarrow\; \dot\theta^2 = \frac{v_0^2}{a^2} - \frac{2g}{a}\left(1-\cos\theta\right) + \Omega^2\sin^2\theta \equiv \Phi(\theta)$$

Quadrature. Separate variables ($d\theta = \sqrt{\Phi}\,dt$, positive root since $\dot\theta(0) = v_0/a > 0$, valid until the first turning point $\Phi = 0$) and integrate from 0 to t:

$$t = \int_0^{\theta(t)} \frac{d\theta'}{\sqrt{\dfrac{v_0^2}{a^2} - \dfrac{2g}{a}\left(1-\cos\theta'\right) + \Omega^2\sin^2\theta'}}$$

Reaction force. From the projection on $\mathbf{e}_n$, $N_n = ma\dot\theta^2 + mg\cos\theta + m\Omega^2a\sin^2\theta$; substituting $\dot\theta^2 = \Phi(\theta)$:

$$N_n(\theta) = \frac{mv_0^2}{a} + mg\left(3\cos\theta - 2\right) + 2m\Omega^2a\sin^2\theta$$

From the projection on $\mathbf{k}_1$:

$$N_z = -2m\Omega a\dot\theta\cos\theta = \mp 2m\Omega a\cos\theta\sqrt{\Phi(\theta)}$$

(upper sign while $\dot\theta > 0$). The total reaction is $\mathbf{N} = N_n\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1) + N_z\,\mathbf{k}_1$.

Staying on the wire. The particle reaches the end $\theta = \pi/2$ only if $\Phi(\pi/2) = v_0^2/a^2 - 2g/a + \Omega^2 \geq 0$; otherwise it oscillates between the two roots of $\Phi(\theta) = 0$. The condition is also sufficient: Φ is a concave quadratic in $\cos\theta$, so on $[0, \pi/2]$ its minimum is at an endpoint, and $\Phi(0) = v_0^2/a^2 > 0$.

(f) Energy equation

> **Pedagogical Justification:** E_1^P is conserved when every force that does work in S_1 derives from a time-independent potential (Slide 29; vault note *Topic 5 - Relative Motion*, Sec. 12.6).

Constant Ω: yes. In S_1: the wire is fixed and smooth, so $\mathbf{N}\cdot\mathbf{v}_1^P = 0$; the Coriolis force never does work, $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ (Slide 31); the weight has the potential $mgy_1 = mga(1-\cos\theta)$; the centrifugal force has the potential $-\tfrac{1}{2}m\Omega^2 r_\perp^2 = -\tfrac{1}{2}m\Omega^2 a^2\sin^2\theta$ (Slide 30), time-independent because Ω is constant. Hence

$$E_1^P = \tfrac{1}{2}ma^2\dot\theta^2 + mga\,(1-\cos\theta) - \tfrac{1}{2}m\Omega^2 a^2\sin^2\theta = \tfrac{1}{2}mv_0^2$$

which is the first integral used in (e). (In S_0 the mechanical energy is not conserved: the reaction N_z is perpendicular to the moving wire but not to the absolute velocity, so the motor that keeps Ω constant does work.)

Variable $\Omega(t)$: no energy integral. Two things change. The Euler force appears, $\mathbf{F}_E = -m\dot\Omega\,\mathbf{j}_1 \times a\sin\theta\,\mathbf{i}_1 = m\dot\Omega a\sin\theta\,\mathbf{k}_1$; it is normal to the plane of the wire, so $\mathbf{F}_E\cdot\mathbf{v}_1^P = 0$ and it does no work in S_1 (it is balanced by N_z). But the centrifugal potential now depends explicitly on time, $V_C = -\tfrac{1}{2}m\Omega(t)^2 a^2\sin^2\theta$, and differentiating E_1^P along the motion (tangential equation with $\Omega(t)$) gives

$$\frac{dE_1^P}{dt} = \frac{\partial V_C}{\partial t} = -m\Omega\dot\Omega a^2\sin^2\theta \neq 0$$

so E_1^P is not a constant of motion and the problem cannot be reduced to a quadrature with the energy equation. The work-energy relation $dE_1^P/dt = \partial V_C/\partial t$ is still valid, but it is not a first integral.

Checks

- **Bifurcation.** For $\Omega^2 < g/a$ gravity dominates and the bottom is the only (stable) equilibrium. At $\Omega_c = \sqrt{g/a}$ the bottom becomes unstable and two stable equilibria $\pm\theta_*$ appear, where the tangential components of weight and centrifugal force balance: $mg\sin\theta_* = m\Omega^2 a \sin\theta_*\cos\theta_*$.
- **Limits.** $\Omega \to 0$: $\Phi = v_0^2/a^2 - 2g(1-\cos\theta)/a$ and $N_n = mv_0^2/a + mg(3\cos\theta - 2)$, the simple-pendulum results; $N_z \to 0$. $\Omega \to \infty$: $\cos\theta_* \to 0$, the equilibria move to the ends of the wire.
- **Coriolis and the reaction.** N_z is exactly the force the wire must exert to cancel the Coriolis force; it changes sign with $\dot\theta$, pushing the particle forward or backward with the rotation.
- **Orthogonality:** $\mathbf{e}_t \cdot \mathbf{e}_n = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$ and $|\mathbf{e}_t| = |\mathbf{e}_n| = 1$.
- **Dimensions:** $[a\sin\theta] = [a(1-\cos\theta)] = \mathrm{m}$, with θ in rad (dimensionless); $[\Omega^2] = [g/a] = \mathrm{s}^{-2}$, so every term of Φ is in s^{-2} and $[t] = \mathrm{rad/s^{-1}} = \mathrm{s}$; $[mv_0^2/a] = [mg] = [m\Omega^2 a] = \mathrm{kg\cdot m\cdot s^{-2}} = \mathrm{N}$; $[m\Omega a\dot\theta] = \mathrm{N}$; $[E_1^P] = [ma^2\dot\theta^2] = \mathrm{J}$.
- **Symbolic check (sympy):** ${}_0R_1$ and $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$; Notes Eq. 3.27 against the double derivative of $\mathbf{r}_1^P$ in $\mathcal{B}_0$; $\mathbf{F}_C$, $\mathbf{F}_{COR}$; the equation of motion and both reaction components from $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{N}$ in S_0; $dE_1^P/dt = 0$; $\Phi(\theta)$ and $N_n(\theta)$; the stability derivatives; and, for $\Omega(t)$, the Euler force, its zero power and $dE_1^P/dt = -m\Omega\dot\Omega a^2\sin^2\theta$. A numerical integration ($g = 9.81$, $a = 1$, $\Omega = 4$, $v_0 = 2$, SI) keeps $\dot\theta^2 - \Phi(\theta)$ below 3×10^{-10}.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 26 in its solution keys, so the results are verified by independent derivation (the symbolic and numerical checks above).

Source: Problems.pdf